



Algorithms and Data Structures

Introduction to practice part of the course

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Common information

- 6 laboratory works
 - Max up to 10 points each
- 3 intermediate tests
 - Max up to 10 points each
- Exam
 - Max up to 10 points



Common information

- Conversion from result score to result mark

Result score (%)	Result grade	Result mark	Explanation
91 – 100	A	Pass	Excellent
84 – 90	B	Pass	Very good
75 – 83	C	Pass	Good
68 – 74	D	Pass	Satisfactory
60 – 67	E	Pass	Satisfactory
0 – 59	F	Fail	Unsatisfactory



Intermediate tests

- One test for each 4 lectures
- 3 test questions + 1 open-ended question
- Variants will be announced later
- Task for test will be shared via link to Yandex Forms and should be filled online
- All fields in test should be filled
- Only the first test submission will be verified



Laboratory works

- Solving problems with different data structures and algorithms
- All laboratory works must be implemented on **C/C++** or **Java**
- Text of all problems is available at
<https://acm.timus.ru/?locale=en>
- **Important** – you should prepare account in Timus system (check slide 9) before first laboratory work classes!
- We have a zero-tolerance policy to any kind of cheating



Laboratory works

- Laboratory works will have following format:
 1. Explanation of possible solution for 1 problem
 2. Students can ask questions related to task and start implementation of a problem (remaining time)
- Mandatory problem should be solved (passed tests on Timus and sent the **report**, check slide 8) before the deadline (check slide 7)
- For homework you will have 1 optional problem to be solved (passed tests on Timus and sent the report)
- **Report** also should be sent before the deadline
- Separate report should be prepared for each problem!
- Homework is **optional**, but it gives greater points
 - For example, in laboratory work #1 for finishing problem #1296 you can get up to 6 points from 10 possible points for this work
 - If you also finishes problem #1005 – you can get up to 10 points (6 + 4) from 10 possible points for this work



Laboratory works plan

- You can check in this table, which problems will be discussed during next practice
- You can find descriptions of problems on Timus
 - E.g. #1296 Hyperjump:
<https://acm.timus.ru/problem.aspx?space=1&num=1296&locale=en>
- Please, read description of a problem before the classes

Classes	Deadline	Type	Related problems	Miscellaneous
Laboratory work #1 25 March	1 April, 22:00	Mandatory problem	#1296 Hyperjump	Up to 6 points
		Optional problem	#1005 Stone Pile	Up to 4 points (not explained)
Laboratory work #2 1 April	8 April, 22:00	Mandatory problem	#1207 Median on the plane	Up to 6 points
		Optional problem	#1401 Gamers	Up to 4 points (not explained)
Laboratory work #3 3 April	10 April, 22:00	Mandatory problem	#1494 Monobilliard	Up to 6 points
		Optional problem	#1726 Visits	Up to 4 points (not explained)
Laboratory work #4 17 April	24 April, 22:00	Mandatory problem	#1521 War games 2	Up to 6 points
		Optional problem	#1650 Billionaires	Up to 4 points (not explained)
Laboratory work #5 24 April	1 May, 22:00	Mandatory problem	#1080 Map coloring	Up to 6 points
		Optional problem	#1806 Mobile Telegraphs	Up to 4 points (not explained)
Laboratory work #6 12 / 14 May	21 May, 22:00	Mandatory problem	#1160 Network	Up to 6 points
		Optional problem	#1162 Currency Exchange	Up to 4 points (not explained)



Laboratory work report

- For each laboratory work a report should be prepared
Use file **template.docx** for format of the report
- All fields of report must be filled
- Reports should contain **explanation** of used algorithm and **computational complexity** of algorithm
- For remote laboratory classes:
 - Report must be prepared in pdf format
 - Reports should be sent to **hduitmo.ads@yandex.ru**
Please, note, email must have proper **subject!**

ADS2026 Lab #N CS Student_Name Student_ID



Working with Timus

1. Prepare account

- <https://acm.timus.ru/register.aspx?locale=en>
- Your name should have following format:
hduads2026_Student-ID

2. Check the guide about working with system

- <https://acm.timus.ru/help.aspx?topic=judge&locale=en>

3. Finish your learning with system

- Check the problem #1000 "A+B Problem"
<https://acm.timus.ru/problem.aspx?space=1&num=1000&locale=en>
- And try to submit your solution for this problem!
<https://acm.timus.ru/submit.aspx?space=1&num=1000&locale=en>

4. If you have language issues, choose "ENG" in top left corner of the page